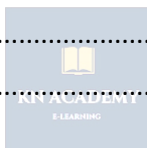


Contents

Average: Formulas and Tips for Quick Calculation.....	1
Important Concepts	3
Key Formulas in Number System	5
Key Rules in Number System.....	6
Important Questions	8
Simple Interest	10
All Concepts in SI.....	12
Percentage	18
Profit & Loss	23
Time & Work	29
Time & Distance	33
Height and Distance	35
Compound Interest	38
Permutation and Combination.....	42
Probability	46
Ratio	49
LCM & HCF	52



Average: Formulas and Tips for Quick Calculation

Key Formulas

Simple Average:

- $\text{Average} = (\text{Sum of all values}) / (\text{Number of values})$

✓ Example:

The ages of five students are 10, 12, 14, 16, and 18 years. Find their average age.

Solution:

$$\text{Average} = 10 + 12 + 14 + 16 + 18 / 5 = 70 / 5 = 14$$

- ♦ Answer: The average age is 14 years.

Weighted Average:

- Weighted Average = (Sum of (weight * value)) / (Sum of weights)
-  **Example:**
A student's exam scores are weighted as follows: Math (40%), Science (30%), and English (30%). If the scores are 85, 90, and 80, find the weighted average score.
- Solution:
- Weighted Average = $(85 \times 0.4) + (90 \times 0.3) + (80 \times 0.3) / (0.4 + 0.3 + 0.3)$
-  **Answer:** The weighted average score is 85.

Average Speed:

- Average Speed = (Total Distance) / (Total Time)

Example:

A car travels **60 km at 30 km/h** and then **60 km at 60 km/h**. Find the average speed.

Solution:

- Time for first trip: $60/30=2$
- Time for second trip: $60/60=1$
- Total distance = $60 + 60 = 120$ km
- Total time = $2 + 1 = 3$ hours



Average Speed = $120/3=40$ km/h

-  **Answer:** The average speed is **40 km/h**.

Successive Discounts:

- **Final Price = Initial Price * (1 - Discount1/100) * (1 - Discount2/100) * ...**
-  **Example:**
A product costs ₹1000 and has two successive discounts of 20% and 10%. Find the final price.
- Solution:
- Price after first discount = $1000 \times (1 - 20/100) = 800$
- Price after second discount = $800 \times (1 - 10/100) = 800 \times 0.9 = 720$
-  **Answer:** The final price is ₹720.

Mixtures:

- (Quantity1 * Concentration1) + (Quantity2 * Concentration2) = (Total Quantity * Final Concentration)

- **Formula (Rule of Alligation):**
- Quantity of Cheaper/Quantity of Expensive=(Expensive Price–Mean Price)/(Mean Price–Cheaper Price)
-  **Example:**
A shopkeeper mixes 10 kg of ₹40/kg sugar with ₹60/kg sugar to get a mixture at ₹50/kg. Find the quantity of ₹60/kg sugar.
- Solution:
Using alligation method:
- (Quantity of ₹40/kg)/(Quantity of ₹60/kg)
- $= (60 - 50) / (50 - 40)$
- $= 10 / 10 = 1:1$
- Thus, 10 kg of ₹60/kg sugar is needed.
-  Answer: 10 kg of costlier sugar is required.

Important Concepts



1. Average When Value is Added or Removed

- **Adding a Value:**
 - **Formula:**
 - $\text{New Average} = (\text{Old Average} * \text{Old Number of Values} + \text{New Value}) / (\text{Old Number of Values} + 1)$
 - **Example:**
 - The average weight of 5 students is 60 kg. If a new student weighing 65 kg joins the group, what is the new average weight?
 - $\text{New Average} = (60 * 5 + 65) / (5 + 1) = 365 / 6 = 60.83 \text{ kg}$
- **Removing a Value:**
 - **Formula:**
 - $\text{New Average} = (\text{Old Average} * \text{Old Number of Values} - \text{Removed Value}) / (\text{Old Number of Values} - 1)$
 - **Example:**

- The average age of 6 employees is 30 years. If an employee aged 35 years leaves the company, what is the new average age?
- New Average = $(30 * 6 - 35) / (6 - 1) = 145 / 5 = 29$ years

2. Average of Odd Numbers

- **Concept:**
 - The average of the first 'n' odd numbers is always 'n'.
 - **Example:**
 - Average of first 5 odd numbers $(1, 3, 5, 7, 9) = (1 + 3 + 5 + 7 + 9) / 5 = 25 / 5 = 5$

3. Average of Even Numbers

- **Concept:**
 - The average of the first 'n' even numbers is 'n + 1'.
 - **Example:**
 - Average of first 4 even numbers $(2, 4, 6, 8) = (2 + 4 + 6 + 8) / 4 = 20 / 4 = 5$

4. Average of Natural Numbers

- **Sum of first 'n' natural numbers:**
 - $\text{Sum} = n * (n + 1) / 2$
- **Average of first 'n' natural numbers:**
 - $\text{Average} = \text{Sum} / n = (n * (n + 1) / 2) / n = (n + 1) / 2$
 - **Example:**
 - Average of first 10 natural numbers = $(10 + 1) / 2 = 5.5$



5. Average if group is replaced

If K members with an average A are replaced by K members with an average B, then:

$$\text{New Average} = \text{Old Average} + \frac{K(B - A)}{\text{Total Members}}$$

✓ **Example:** A team of 10 members has an average age of 30 years. If 2 members aged 40 are replaced by 2 members aged 25, find the new average.

Solution:

$$\text{New Average} = 30 + \frac{2(25 - 40)}{10} = 30 + \frac{-30}{10} = 30 - 3 = 27$$

♦ **Answer: 27 years**

6. Average when mistake is created

If a wrong value X was included in the calculation and should be replaced with Y , then:

$$\text{New Average} = \text{Old Average} + \frac{(Y - X)}{\text{Total Items}}$$

✓ **Example:** The average of 5 numbers was mistakenly calculated as 50, but one number 60 was mistakenly taken as 40. Find the correct average.

Solution:

$$\text{Correct Average} = 50 + \frac{(60 - 40)}{5} = 50 + \frac{20}{5} = 50 + 4 = 54$$

♦ **Answer: 54**

Key Formulas in Number System

1. Sum of First 'n' Natural Numbers:

- Sum = $n(n+1)/2$



2. Sum of Squares of First 'n' Natural Numbers:

- Sum = $n(n+1)(2n+1)/6$

3. Sum of Cubes of First 'n' Natural Numbers:

- Sum = $[n(n+1)/2]^2$

4. Euclid's Division Lemma:

- For any two integers 'a' and 'b', there exist unique integers 'q' and 'r' such that:
 - $a = bq + r$
 - where $0 \leq r < b$

5. Relationship between LCM and HCF:

- For any two integers 'a' and 'b':
 - $\text{LCM}(a, b) * \text{HCF}(a, b) = a * b$
- ✓ **Example:** Find LCM and HCF of 12 and 18
- Step 1: Find HCF**
Using prime factorization:
 - $12 = 2^2 \times 3, 18 = 2 \times 3^2$

- $HCF = 2^1 \times 3^1 = 6$
- **Step 2: Find LCM**
- $LCM = 2^2 \times 3^2 = 36$
- **Step 3: Verify Relationship**
- $LCM \times HCF = 36 \times 6 = 216$
-

Important Algebraic Identities:

- $(a + b)^2 = a^2 + 2ab + b^2$
- $(a - b)^2 = a^2 - 2ab + b^2$
- $(a + b)(a - b) = a^2 - b^2$
- $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$
- $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$
- $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$

Fermat Theorem

Fermat's Little Theorem is a fundamental result in number theory, used in modular arithmetic and cryptography.

🚩 Theorem Statement:

If p is a prime number, and a is an integer not divisible by p , then:

$$a^{p-1} \equiv 1 \pmod{p}$$

This means that when a^{p-1} is divided by p , the remainder is 1.

🚩 Alternate Form (for Exponent Reduction):

For any integer a ,

$$a^p \equiv a \pmod{p}$$

This is useful in modular exponentiation when reducing large powers.

Key Rules in Number System

1. Divisibility Rules: These rules help determine if a number is divisible by another number without performing actual division.

- **Divisible by 2:** If the last digit is even (0, 2, 4, 6, 8).
- **Divisible by 3:** If the sum of the digits is divisible by 3.

- **Divisible by 4:** If the last two digits are divisible by 4.
- **Divisible by 5:** If the last digit is 0 or 5.
- **Divisible by 6:** If the number is divisible by both 2 and 3.
- **Divisible by 7:** Double the last digit and subtract it from the remaining number. Repeat until you get a single-digit number. If this number is 0 or 7, the original number is divisible by 7.
- **Divisible by 8:** If the last three digits are divisible by 8.
- **Divisible by 9:** If the sum of the digits is divisible by 9.
- **Divisible by 10:** If the last digit is 0.
- **Divisible by 11:** The difference between the sum of digits at odd places and the sum of digits at even places is 0 or divisible by 11.

2. Prime Numbers:

- A prime number is a natural number greater than 1 that has only two divisors: 1 and itself.
- The first few prime numbers are: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, ...

3. Composite Numbers:

- A composite number is a natural number greater than 1 that has more than two divisors.
- Examples: 4, 6, 8, 9, 10, 12, ...



4. Co-prime Numbers:

- Two numbers are co-prime if their greatest common divisor (GCD) is 1.
- Example: 8 and 15 are co-prime because their GCD is 1.

5. LCM (Least Common Multiple):

- The smallest number that is a multiple of two or more given numbers.

6. HCF (Highest Common Factor) or GCD (Greatest Common Divisor):

- The largest number that divides two or more given numbers without leaving a remainder.

7. Euclid's Division Lemma:

- For any two integers 'a' and 'b', there exist unique integers 'q' and 'r' such that:
 - $a = bq + r$
 - where $0 \leq r < b$

8. Euclid's Division Algorithm:

- Used to find the HCF of two numbers based on Euclid's Division Lemma.

9. Number Line:

- A visual representation of numbers on a line, which helps in understanding inequalities and operations.

10. Unit Digit Patterns:

- Recognizing repeating patterns in the unit digits of powers of numbers (e.g., $2^1, 2^2, 2^3, \dots$)

Important Questions

Q1: Find the remainder when 7^{103} is divided by 5.

Solution:

Using Fermat's Theorem, if p is a prime number and a is not divisible by p , then:

$$a^{p-1} \equiv 1 \pmod{p}$$

For $7^{103} \pmod{5}$:

- Since $7 \equiv 2 \pmod{5}$, we rewrite the exponentiation:

$$7^{103} \equiv 2^{103} \pmod{5}$$

- The powers of 2 mod 5 cycle every 4:

$$2^1 \equiv 2, \quad 2^2 \equiv 4, \quad 2^3 \equiv 3, \quad 2^4 \equiv 1 \pmod{5}$$

- Since $103 \pmod{4} = 3$, we take the 3rd power in the cycle:

$$2^{103} \equiv 3 \pmod{5}$$

✓ Answer: 3

Q2: Find the last two digits of 13^{45} .

Solution:

We need to find $13^{45} \pmod{100}$. Using Euler's Theorem,

$$13^{\phi(100)} \equiv 1 \pmod{100}, \quad \text{where } \phi(100) = 40$$

Thus,

$$13^{45} = 13^{40} \times 13^5 \equiv 1 \times 13^5 \pmod{100}$$

Finding $13^5 \pmod{100}$:

$$13^2 = 169 \equiv 69 \pmod{100}$$

$$13^4 = (13^2)^2 = 69^2 = 4761 \equiv 61 \pmod{100}$$

$$13^5 = 13^4 \times 13 = 61 \times 13 = 793 \equiv 93 \pmod{100}$$

✓ Answer: 93

Q3: What is the smallest number that leaves remainders 1, 2, and 3 when divided by 5, 6, and 7 respectively?

Solution:

We need a number N such that:

$$N \equiv 1 \pmod{5}$$

$$N \equiv 2 \pmod{6}$$

$$N \equiv 3 \pmod{7}$$

Using Chinese Remainder Theorem, we express N as:

$$N = 5k + 1$$

$$5k + 1 \equiv 2 \pmod{6} \Rightarrow 5k \equiv 1 \pmod{6}$$

Solving $5k = 6m + 1$, we find $k = 5$ works (as $5 \times 5 = 25 \equiv 1 \pmod{6}$).

Thus,

$$N = 5(5) + 1 = 26$$

Checking $N \equiv 3 \pmod{7}$:

$$26 \pmod{7} = 26 - 3(7) = 26 - 21 = 5 \neq 3$$

So, trying $N = 26 + \text{LCM}(5, 6) = 56$,

$$56 \equiv 1 \pmod{5}, \quad 56 \equiv 2 \pmod{6}, \quad 56 \equiv 3 \pmod{7}$$

✅ Answer: 56



Q4: Find the number of zeroes at the end of $100!$.

Solution:

The number of trailing zeroes in $n!$ is given by:

$$\sum \left\lfloor \frac{n}{5^k} \right\rfloor$$

For $100!$:

$$\begin{aligned} & \left\lfloor \frac{100}{5} \right\rfloor + \left\lfloor \frac{100}{25} \right\rfloor + \left\lfloor \frac{100}{125} \right\rfloor \\ &= 20 + 4 + 0 = 24 \end{aligned}$$

✅ Answer: 24

Q5: Find the remainder when 3^{100} is divided by 101.

Solution:

Using Fermat's Theorem, since 101 is prime:

$$3^{100} \equiv 1 \pmod{101}$$

✓ Answer: 1

Simple Interest

Formulas

1. Basic Formula of Simple Interest

$$SI = \frac{P \times R \times T}{100}$$

Where:

- SI = Simple Interest
- P = Principal (Initial Amount)
- R = Rate of Interest per annum (%)
- T = Time (in years)

2. Amount Calculation

$$A = P + SI$$

Where:

- A = Total Amount
- P = Principal
- SI = Simple Interest

Finding Values

1. Find Principal (P)

$$P = \frac{SI \times 100}{R \times T}$$

2. Find Rate of Interest (R)

$$R = \frac{SI \times 100}{P \times T}$$

3. Find Time (T)

$$T = \frac{SI \times 100}{P \times R}$$

4. Special Cases & Tricks

✓ When Interest is given for months:

$$T = \frac{\text{Number of months}}{12}$$

✓ When Interest is given for days:

$$T = \frac{\text{Number of days}}{365}$$

Special Cases

4. Special Cases & Tricks

✓ When Interest is given for months:

$$T = \frac{\text{Number of months}}{12}$$

✓ When Interest is given for days:

$$T = \frac{\text{Number of days}}{365}$$

(Use 365 for normal years and 366 for leap years)

✓ When Rate is given for different periods:

- If rate is given half-yearly, divide R by 2.
- If rate is given quarterly, divide R by 4.

✓ Doubling the Principal using SI Formula:

$$T = \frac{100}{R}$$

(For amount to become double, $SI = P$, so $P + SI = 2P$)

✓ Tripling the Principal using SI Formula:

$$T = \frac{200}{R}$$

(For amount to become triple, $SI = 2P$, so $P + SI = 3P$)

All Concepts in SI

1. Simple Interest (SI)

- Simple Interest is the interest calculated only on the **principal amount** (initial investment or loan) for a fixed period at a fixed rate.
 - Formula: $SI = P \times R \times T / 100$
 - P = Principal
 - R = Rate of interest per annum (%)
 - T = Time (years)
-

2. Principal (P)

- The original sum of money deposited, borrowed, or invested.
 - Example: If you invest ₹5000, then ₹5000 is the **principal**.
-

3. Interest

- The extra amount paid by a borrower to a lender for using money over a certain period.
 - Interest can be **Simple Interest (SI)** or **Compound Interest (CI)**.
-

4. Rate of Interest (R)

- The percentage at which interest is calculated on the principal.
 - It is usually given **annually** (per year).
 - Example: If the rate is **10% per annum**, it means that ₹10 is earned per ₹100 per year.
-

5. Time (T)

- The period for which the money is borrowed, invested, or deposited.
 - It is usually measured in **years** but can be converted to **months** or **days**.
 - **Months:** $T = \text{Months} / 12$
 - **Days:** $T = \text{Days} / 365$ (or **366 in leap year**)
-

6. Amount (A)

- The total money received or paid after a certain period, including the principal and interest.
- Formula: $A = P + SI$

- **Example:** If $P = ₹5000$ and $SI = ₹1000$, then $A = 5000 + 1000 = 6000$
-

7. Loan

- Money borrowed from a lender (bank, person, etc.) that must be repaid with interest.
-

8. Borrower

- The person or organization that **takes a loan** and agrees to repay it with interest.
-

9. Lender

- The person, bank, or financial institution that **gives a loan** and earns interest from it.
-

10. Profit / Earnings

- The total extra amount earned as interest from an investment.
 - In simple interest, **profit = SI**.
-

11. Doubling Time

- The time required for the principal to **double** in value under simple interest.
 - Formula: $T = 100/R$
 - Example: If $R = 5\%$, then $T = 100/5 = 20$ years
 - It means that at **5% SI per year**, the amount will double in **20 years**.
-

12. Tripling Time

- The time required for the principal to **triple** in value under simple interest.
 - Formula: $T = 200/R$
-

13. Half-Yearly Interest

- If interest is calculated **twice a year**, the rate is **divided by 2**, and time is **doubled**.
 - New rate = $R/2$
 - New time = $2T$
-

14. Quarterly Interest

- If interest is calculated **four times a year**, the rate is **divided by 4**, and time is **multiplied by 4**.
 - New rate = $R/4$
 - New time = $4T$

15. Simple Interest vs. Compound Interest

Feature	Simple Interest (SI)	Compound Interest (CI)
Interest Calculation	Only on Principal (P)	On Principal + Interest
Formula	$SI = P \times R \times T / 100$	$A = P(1 + R/100)^T$
Growth	Linear (Constant Increase)	Exponential (Grows Faster)
Example	₹1000 at 10% for 3 years = ₹300	₹1000 at 10% for 3 years = ₹331

Questions

● Question 1: Time Calculation in a Complex Scenario

A sum of ₹12,000 is invested at 5% per annum simple interest. After some years, the total interest becomes ₹3,600. Find the time in years.

Solution:

We use the formula:

$$SI = \frac{P \times R \times T}{100}$$

Substituting the given values:

$$3600 = \frac{12000 \times 5 \times T}{100}$$

$$3600 = \frac{60000T}{100}$$

$$3600 \times 100 = 60000T$$

$$T = \frac{360000}{60000} = 6 \text{ years}$$

✅ Answer: 6 years

● Question 2: Finding Principal with Changing Interest Rate

A person earns ₹10,500 as simple interest in 5 years. The interest rate was 4% for the first 2 years, and 6% for the next 3 years. Find the principal amount.

Solution:

Split the SI into two parts:

1 For the first 2 years (4% rate):

$$SI_1 = \frac{P \times 4 \times 2}{100} = \frac{8P}{100} = \frac{8P}{100}$$

2 For the next 3 years (6% rate):

$$SI_2 = \frac{P \times 6 \times 3}{100} = \frac{18P}{100}$$

Total SI:

$$SI_1 + SI_2 = 10,500$$

$$\frac{8P}{100} + \frac{18P}{100} = 10,500$$

$$\frac{26P}{100} = 10,500$$

$$P = \frac{10,500 \times 100}{26} = \frac{1,050,000}{26} = 40,384.62$$

✓ Answer: ₹40,384.62

● Question 3: Comparing Two Loans

A person borrowed ₹X from a bank at 8% per annum simple interest for 3 years. Another person borrowed ₹(X + 5000) at 6% per annum for the same period. If the interest amounts are equal, find the value of X.

Solution:

Using the formula for SI:

1 For the first person:

$$SI_1 = \frac{X \times 8 \times 3}{100} = \frac{24X}{100}$$

2 For the second person:

$$SI_2 = \frac{(X + 5000) \times 6 \times 3}{100} = \frac{18(X + 5000)}{100}$$

Since both interests are equal:

$$\frac{24X}{100} = \frac{18(X + 5000)}{100}$$

Cancel 100 from both sides:

$$24X = 18(X + 5000)$$

$$24X = 18X + 90,000$$

$$6X = 90,000$$

$$X = 15,000$$

✓ Answer: ₹15,000

● Question 4: Time to Double the Money

In how many years will ₹20,000 amount to ₹40,000 at an interest rate of 10% per annum under simple interest?

Solution:

Total amount after T years:

$$A = P + SI$$

$$40,000 = 20,000 + \frac{20,000 \times 10 \times T}{100}$$

$$40,000 = 20,000 + \frac{200,000T}{100}$$

$$40,000 - 20,000 = 2,000T$$

$$20,000 = 2,000T$$

$$T = \frac{20,000}{2,000} = 10 \text{ years}$$

✓ Answer: 10 years

● Question 5: Ratio-Based SI Problem

A sum of money is invested at 6% per annum and earns ₹900 in 2 years. If the same sum was invested at R% per annum for 5 years, the interest earned would be ₹3750. Find the value of R.

Solution:

1 Find the principal using first case:

$$SI = \frac{P \times 6 \times 2}{100}$$
$$900 = \frac{12P}{100}$$
$$P = \frac{900 \times 100}{12} = 7500$$

2 Use the principal to find R:

$$SI = \frac{P \times R \times 5}{100}$$
$$3750 = \frac{7500 \times R \times 5}{100}$$
$$3750 \times 100 = 7500 \times 5R$$
$$375000 = 37500R$$
$$R = \frac{375000}{37500} = 10\%$$

✓ Answer: 10%



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Percentage

1. Definitions

✓ Percentage

- A percentage is a **fraction or ratio** expressed as a part of 100.
- Symbol: %
- Example: 50% means 50 out of 100 or $\frac{50}{100} = 0.5$

✓ Percentage Formula

$$\text{Percentage} = \left(\frac{\text{Value}}{\text{Total Value}} \right) \times 100$$

Example: If 30 students out of 50 passed the exam, the pass percentage is:

$$\left(\frac{30}{50} \right) \times 100 = 60\%$$

Conversion

Fraction	Decimal	Percentage
$\frac{1}{2}$	0.5	50%
$\frac{1}{3}$	0.333	33.33%
$\frac{1}{4}$	0.25	25%
$\frac{3}{4}$	0.75	75%

To convert:

- **Fraction to Percentage** → Multiply by 100
- **Decimal to Percentage** → Multiply by 100
- **Percentage to Decimal** → Divide by 100

Percentage Increase/Decrease

✓ Percentage Increase/Decrease

1. Percentage Increase:

$$\text{Increase} = \left(\frac{\text{New Value} - \text{Old Value}}{\text{Old Value}} \right) \times 100$$

Example: If a price increases from ₹200 to ₹250:

$$\left(\frac{250 - 200}{200} \right) \times 100 = 25\% \text{ increase}$$

2. Percentage Decrease:

$$\text{Decrease} = \left(\frac{\text{Old Value} - \text{New Value}}{\text{Old Value}} \right) \times 100$$

Example: If a price drops from ₹500 to ₹400:

$$\left(\frac{500 - 400}{500} \right) \times 100 = 20\% \text{ decrease}$$

Important Formulas



✓ Finding a Percentage of a Number

$$X\% \text{ of } Y = \frac{X}{100} \times Y$$

Example: 20% of 500

$$\frac{20}{100} \times 500 = 100$$

✓ Finding What Percentage One Number is of Another

$$\text{Percentage} = \left(\frac{\text{Part}}{\text{Whole}} \right) \times 100$$

Example: What percentage is 40 of 200?

$$\left(\frac{40}{200} \right) \times 100 = 20\%$$

✓ Percentage Change After Multiple Increases/Decreases

If a value increases by $x\%$ and then by $y\%$, the total percentage change is:

$$\text{Total Change} = x + y + \frac{x \times y}{100}$$

Example: If a price increases by 20%, then by 30%:

$$\text{Total Increase} = 20 + 30 + \frac{20 \times 30}{100} = 56\%$$

For a successive decrease:

$$\text{Total Decrease} = -x - y + \frac{x \times y}{100}$$

Finding the original value

1. If a number increases by $X\%$ and becomes Y , find the original value:

$$\text{Original Value} = \frac{Y \times 100}{100 + X}$$

Example: If a price increases by 25% and becomes ₹250, the original price was:

$$\frac{250 \times 100}{100 + 25} = \frac{25000}{125} = 200$$

2. If a number decreases by $X\%$ and becomes Y , find the original value:

$$\text{Original Value} = \frac{Y \times 100}{100 - X}$$

✔ Percentage Profit & Loss

1. Profit Percentage:

$$\text{Profit \%} = \left(\frac{\text{Profit}}{\text{Cost Price}} \right) \times 100$$

2. Loss Percentage:

$$\text{Loss \%} = \left(\frac{\text{Loss}}{\text{Cost Price}} \right) \times 100$$

Shortcut Formula

✔ Successive Increase/Decrease Shortcut:

- If an item increases by 10% twice, total increase = $10 + 10 + \frac{10 \times 10}{100} = 21\%$
- If an item decreases by 20% and then 10%, total decrease = $20 + 10 - \frac{20 \times 10}{100} = 28\%$

✔ Reverse Percentage Trick

- If a price increases by $X\%$ and the final value is Y , the original value is:

$$\text{Original Value} = \frac{Y \times 100}{100 + X}$$

- If a price decreases by $X\%$ and the final value is Y , the original value is:

$$\text{Original Value} = \frac{Y \times 100}{100 - X}$$

Questions

Question:

A shopkeeper gives a 20% discount on an item and still makes a 25% profit. If the cost price of the item is ₹600, what was the original marked price?

Solution:

Let the marked price be M and the selling price be S .

- After a 20% discount, the selling price is:

$$S = 80\% \text{ of } M = \frac{80}{100} \times M = 0.8M$$

- Since the shopkeeper makes a 25% profit, the selling price is:

$$S = 125\% \text{ of cost price} = \frac{125}{100} \times 600 = 750$$

- Now equating both selling price expressions:

$$0.8M = 750$$

- Solving for M :

$$M = \frac{750}{0.8} = 937.50$$

✓ Answer: The marked price of the item was ₹937.50.

◆ Question 2: Successive Percentage Change

Question:

The population of a city increases by 10% in the first year, then decreases by 15% in the second year, and then increases by 20% in the third year. What is the net percentage change over the three years?

Solution:

Using the successive percentage formula:

1st and 2nd year:

$$10 + (-15) + \frac{10 \times (-15)}{100} = -6.5\%$$

Now applying -6.5% change with +20% for the 3rd year:

$$-6.5 + 20 + \frac{-6.5 \times 20}{100} = 12.7\%$$

✓ Answer: Net increase of 12.7%

◆ Question 3: A Trick in Elections

Question:

In an election, a candidate gets 56% of the votes and wins by 1680 votes. What is the total number of votes cast?

Solution:

- Let the total votes be X .
- The losing candidate gets $(100 - 56) = 44\%$ of the votes.
- The difference is 12% of total votes, i.e.,

$$\frac{12}{100} \times X = 1680$$
$$X = \frac{1680 \times 100}{12} = 14000$$

✓ Answer: 14,000 votes

◆ Question 4: Percentage of Mixtures

Question:

A 50-liter mixture of milk and water contains 30% water. How much more water must be added to make the mixture 50% water?

Solution:

- Water in the original mixture = 30% of 50 = 15 liters
- Let x liters of water be added.
- New total volume = $(50 + x)$ liters
- New water percentage equation:

$$\frac{15 + x}{50 + x} = \frac{50}{100}$$

Cross multiply:

$$(15 + x) \times 2 = 50 + x$$

$$30 + 2x = 50 + x$$

$$2x - x = 50 - 30$$

$$x = 20$$

✓ Answer: 20 liters of water

Question:

A man's salary is reduced by 20%, and later his salary is increased by 25%. What is the overall percentage change in salary?

Solution:

Using the successive percentage change formula:

$$\text{Total Percentage Change} = A + B + \frac{A \times B}{100}$$

where:

- $A = -20\%$ (salary reduction)
- $B = 25\%$ (salary increase)

Substituting the values:

$$\begin{aligned} &= (-20) + 25 + \frac{(-20) \times 25}{100} \\ &= -20 + 25 - 5 \\ &= 0\% \end{aligned}$$

◆ Question 6: Percentage in Work**Question:**

A contractor employs 50 workers to complete a project in 40 days. If 20% of the workers leave after 10 days, how many extra days will be required to finish the work?

Solution:

- Total work = $50 \times 40 = 2000$ worker-days
- In 10 days, $50 \times 10 = 500$ worker-days are completed.
- Remaining work = $2000 - 500 = 1500$ worker-days
- Now, 20% workers leave, so new workforce = 80% of 50 = 40 workers
- Time required with 40 workers:

$$\frac{1500}{40} = 37.5 \text{ days}$$

✔ Answer: 38 days

Profit & Loss

✔ Cost Price (CP)

- The **original price** at which an item is **bought**.
- Example: If you buy a phone for ₹10,000, then **CP = ₹10,000**.

✓ Selling Price (SP)

- The **final price** at which an item is **sold**.
 - Example: If you sell the phone for ₹12,000, then **SP = ₹12,000**.
-

✓ Profit (Gain)

- If **Selling Price (SP) > Cost Price (CP)**, the seller makes a **profit**.
 - Formula: Profit=SP-CP
 - Example: If **CP = ₹500** and **SP = ₹600**, then Profit=600-500=₹100
-

✓ Loss

- If **Selling Price (SP) < Cost Price (CP)**, the seller incurs a **loss**.
 - Formula: Loss=CP-SP
 - Example: If **CP = ₹800** and **SP = ₹700**, then Loss=800-700=₹100
-

2. Important Formulas



✓ Profit Percentage (Profit %)

$$\text{Profit \%} = \left(\frac{\text{Profit}}{\text{CP}} \right) \times 100$$

Example: If **CP = ₹1000** and **Profit = ₹200**,

$$\text{Profit \%} = \left(\frac{200}{1000} \right) \times 100 = 20\%$$

✓ Loss Percentage (Loss %)

$$\text{Loss \%} = \left(\frac{\text{Loss}}{\text{CP}} \right) \times 100$$

Example: If **CP = ₹5000** and **Loss = ₹500**,

$$\text{Loss \%} = \left(\frac{500}{5000} \right) \times 100 = 10\%$$

✓ Selling Price (SP) Formulas

1. When Profit % is given:

$$SP = CP \times \left(1 + \frac{\text{Profit \%}}{100}\right)$$

2. When Loss % is given:

$$SP = CP \times \left(1 - \frac{\text{Loss \%}}{100}\right)$$

Example: If CP = ₹1000 and Profit = 20%, then

$$SP = 1000 \times \left(1 + \frac{20}{100}\right) = 1000 \times 1.2 = ₹1200$$

✓ Cost Price (CP) Formulas

1. When Profit % is given:

$$CP = \frac{SP}{1 + \frac{\text{Profit \%}}{100}}$$

2. When Loss % is given:

$$CP = \frac{SP}{1 - \frac{\text{Loss \%}}{100}}$$

Example: If SP = ₹1200 and Profit = 20%, then

$$CP = \frac{1200}{1.2} = ₹1000$$

✓ Discount Formula

1. Discount = Marked Price (MP) - Selling Price (SP)

2. Discount %

$$\text{Discount \%} = \left(\frac{\text{Discount}}{\text{Marked Price}}\right) \times 100$$

Example: If a watch has MP = ₹5000 and a Discount of ₹1000 is given,

$$\text{Discount \%} = \left(\frac{1000}{5000}\right) \times 100 = 20\%$$

✓ Successive Profit & Loss

If an item is increased by **x%** and then by **y%**, the total increase is:

Example: If a price increases by **20%** and then by **10%**,

Total Increase = $20 + 10 + (20 \times 10 / 100) = 32\%$

For **Successive Decrease**:

Total Decrease = $-x - y + (x \times y / 100)$

Total Decrease = $-x - y + (x \times y / 100)$

3. Concepts and Tricks

✓ **Shortcut for Quick Calculation:**

- 10% increase → Multiply by 1.1
- 20% increase → Multiply by 1.2
- 10% decrease → Multiply by 0.9
- 20% decrease → Multiply by 0.8

✓ **Relationship Between CP, SP, and Profit/Loss**

$SP = CP + \text{Profit} = CP - \text{Loss}$

✓ **If SP and CP Difference is Given:**

- If $SP > CP \rightarrow \text{Profit}$
- If $SP < CP \rightarrow \text{Loss}$

✓ **Breaking Even:**

- If $SP = CP$, no profit, no loss.

✓ **Double Profit Rule:**

- If an item's price **doubles (100% increase)**, the new profit % is **100+old profit**



Basic Examples

Example 1: Profit Calculation

A shopkeeper buys a pen for ₹40 and sells it for ₹50. Find the profit and profit %.

Solution:

$$\text{Profit} = SP - CP = 50 - 40 = ₹10$$

$$\text{Profit \%} = \left(\frac{10}{40} \right) \times 100 = 25\%$$

Example 2: Loss Calculation

A person bought a TV for ₹25,000 and sold it for ₹20,000. Find the loss and loss %.

Solution:

$$\text{Loss} = CP - SP = 25000 - 20000 = ₹5000$$

$$\text{Loss \%} = \left(\frac{5000}{25000} \right) \times 100 = 20\%$$

Example 3: Finding CP

A person sold an item for ₹600 and made a profit of 25%. Find the Cost Price.

Solution:

$$CP = \frac{SP}{1 + \frac{\text{Profit\%}}{100}} = \frac{600}{1.25} = ₹480$$

- ◆ **Question:** A shopkeeper increases the price of an item by 40% and then offers a discount of 25%.

What is the overall percentage gain or loss?

✓ **Solution:**

Let the original price be ₹100.

- **After 40% increase:** New Price = $100 + (40\% \text{ of } 100) = ₹140$
- **After 25% discount:** Selling Price = $140 - (25\% \text{ of } 140)$
 $= 140 - 35 = ₹105$
- **Profit** = $105 - 100 = ₹5$
- **Profit %** = $(5/100) \times 100 = 5\%$

✚ **Answer:** 5% profit

♦ **Question:** A trader makes **20% profit** by selling an article for ₹1,800. If he had sold it for ₹1,680, what would have been his profit or loss percentage?

✔ **Solution:**

Let the **cost price (CP)** be x .

- Selling Price (SP) at **20% profit** = $x \times (1 + 20/100) = 1.2x$
 $1.2x = 1800$
 $x = 1800 / 1.2 = \text{₹}1500$
- If he sells at ₹1,680:
Profit = $1680 - 1500 = \text{₹}180$
Profit % = $(180 / 1500) \times 100 = 12\%$

✚ **Answer:** 12% profit

♦ **Question:** A shopkeeper offers **three successive discounts** of 10%, 20%, and 30% on a product marked ₹5,000. What is the final selling price?

✔ **Solution:**

- **First discount:** 10% of 5000 = 500 → New Price = $5000 - 500 = \text{₹}4500$
- **Second discount:** 20% of 4500 = 900 → New Price = $4500 - 900 = \text{₹}3600$
- **Third discount:** 30% of 3600 = 1080 → New Price = $3600 - 1080 = \text{₹}2,520$

✚ **Answer:** ₹2,520

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♦ **Question:** A shopkeeper marks his goods **40% above the cost price** and then gives a discount of 25%. Find his actual profit percentage.

✔ **Solution:**

Let the **cost price (CP)** = ₹100

- **Marked Price (MP)** = $100 + 40\% \text{ of } 100 = \text{₹}140$
- **After 25% discount:** Selling Price = $140 - (25\% \text{ of } 140)$
 $= 140 - 35 = \text{₹}105$
- **Profit %** = $[(105 - 100) / 100] \times 100 = 5\%$

✚ **Answer:** 5% profit

◆ **Question:** A person sold a table at **20% profit**. If he had sold it for ₹200 more, his profit would have been **40%**. Find the cost price of the table.

✔ **Solution:**

Let CP = x .

- **20% profit:** $SP = x \times (1.2)$
- **40% profit:** $SP = x \times (1.4)$
- Difference = ₹200
 $\Rightarrow 1.4x - 1.2x = 200$
 $\Rightarrow 0.2x = 200$
 $\Rightarrow x = \text{₹}1,000$

🚩 **Answer:** ₹1,000

Time & Work

1. Definitions

✔ Work

- Work refers to a task that must be completed.
- It is usually measured in terms of the number of days or hours required to complete it.

✔ Time Taken

- The total number of days or hours required to complete a work.
- Example: If a person can paint a house in 10 days, the time taken is 10 days.

✔ Efficiency

- Efficiency refers to the rate at which a person or machine completes work.
- If a person completes $\frac{1}{10}$ th of the work per day, his efficiency is 10% per day.

✔ Work Done

- Work done is the fraction of work completed in a given time.
- Formula:
$$\text{Work Done} = \text{Efficiency} \times \text{Time}$$
- Example: If a worker completes $\frac{1}{5}$ th of the work in a day, then in 2 days, he completes $\frac{2}{5}$ th of the work.

Important Formulas

✔ Basic Time and Work Formula

$$\text{Work Done} = \text{Time Taken} \times \text{Efficiency}$$

✔ Relation Between Work, Time, and Number of Workers

More workers $\Rightarrow$ Less time required

Less workers $\Rightarrow$ More time required

Formula:

$$\text{Work} = \text{Number of People} \times \text{Time Taken} \times \text{Efficiency}$$

✔ Work Done by One Person

If A alone takes X days to complete a task, then:

$$A's \text{ Work Rate} = \frac{1}{X} \text{ work per day}$$

Example: If A can finish a task in 5 days, then A does $\frac{1}{5}$ of the work per day.

✔ Total Work Formula

If A completes a task in X days, then:

$$\text{Total Work} = A's \text{ work per day} \times \text{Total Days} = 1$$

Example: If A can complete the work in 10 days,

$$\text{Total Work} = 1 \text{ (full work)}$$

✔ Work Done Together

If A alone completes work in X days and B alone completes the same work in Y days, then:

$$\text{Work done by A and B in one day} = \left(\frac{1}{X} + \frac{1}{Y} \right)$$

$$\text{Total Time Taken} = \frac{XY}{X + Y}$$

Example:

- A alone finishes in 6 days and B alone in 12 days.
 - Work done together in one day =
$$\frac{1}{6} + \frac{1}{12} = \frac{2}{12} + \frac{1}{12} = \frac{3}{12} = \frac{1}{4}$$
 - So, A and B together finish in 4 days.
-

✔ Inverse Proportionality (More People, Less Time)

If the number of workers increases, the time required decreases proportionally.

Formula:

$$M_1 \times D_1 = M_2 \times D_2$$

Where:

- M_1 and M_2 are the number of workers in the first and second scenario.
- D_1 and D_2 are the days required.

Example:

If 10 workers complete a task in 20 days, then how many days will 5 workers take?

$$10 \times 20 = 5 \times D$$

$$D = \frac{10 \times 20}{5} = 40 \text{ days}$$

✓ Efficiency Shortcut

$$\text{Efficiency} = \frac{100}{\text{Total Time Taken}}$$

Example: If A completes the work in 10 days, his efficiency is:

$$\frac{100}{10} = 10\%$$

This means A completes 10% of the work per day.

✓ Work and Wages Relation

$$\text{Wages} \propto \text{Work Done}$$

If A and B work together and earn ₹600, their share is divided based on the work done.

If A does 3/5th of the work and B does 2/5th:

- A gets $(3/5) \times 600 = ₹360$
- B gets $(2/5) \times 600 = ₹240$



Fast Calculations

✓ 1. Shortcut for Two Workers Working Together

If A completes work in X days and B in Y days,

$$\text{Total Work Time} = \frac{XY}{X + Y}$$

Use this to quickly calculate when two people work together.

✓ 2. If A is Twice as Fast as B

- If A alone takes 10 days, and A is twice as fast as B,
 - B takes 20 days.
 - Together they take:

$$\frac{10 \times 20}{10 + 20} = \frac{200}{30} = 6.67 \text{ days}$$

✓ 3. If Efficiency is Given in Ratio

- If A and B's efficiency is in 3:2,
 - A does 3/5th of the work
 - B does 2/5th
 - Total time = (Total Work / Sum of Efficiency)

✓ 4. If A Takes x% More/Less Time Than B

- If A takes 25% more time than B,
 - A's time = $1.25 \times \text{B's time}$
- If A takes 20% less time than B,
 - A's time = $0.8 \times \text{B's time}$

✓ 5. Shortcut for Pipes & Cisterns

- If Pipe A fills in X minutes and Pipe B empties in Y minutes,

$$\text{Time to Fill} = \frac{XY}{Y - X}$$

Example Problems

Example 1: Work Together

A can complete a task in 12 days, and B can complete the same task in 18 days. How long will they take together?

Solution:

$$\text{Total Work} = \frac{12 \times 18}{12 + 18} = \frac{216}{30} = 7.2 \text{ days}$$

Example 2: Worker Ratio

If 8 workers can complete a job in 24 days, how many days will 6 workers take?

Solution:

Using the formula:

$$M_1 \times D_1 = M_2 \times D_2$$

$$8 \times 24 = 6 \times D$$

$$D = \frac{8 \times 24}{6} = 32 \text{ days}$$

Example 3: Work Sharing

A and B together complete a work for ₹1500. A alone completes the work in 6 days, B alone in 9 days.

What is B's share?

Solution:

Work done by A and B in one day:

$$\frac{1}{6} + \frac{1}{9} = \frac{3}{18} + \frac{2}{18} = \frac{5}{18}$$

A's work share = $3/5$, B's share = $2/5$

$$B's \text{ earnings} = \frac{2}{5} \times 1500 = ₹600$$

Time & Distance

✓ Distance (D)

- The **total length of the path** traveled by an object.
- It is usually measured in **kilometers (km) or meters (m)**.

✓ Speed (S)

- The **rate at which an object covers distance**.
- It is measured in **km/hr or m/sec**.

✓ Time (T)

- The **total duration taken** to travel a certain distance.
- Measured in **hours, minutes, or seconds**.

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Basic Formulas :

2. Basic Formula

The most fundamental formula in Time, Speed & Distance problems:

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

$$\text{Time} = \frac{\text{Distance}}{\text{Speed}}$$

$$\text{Distance} = \text{Speed} \times \text{Time}$$

Conversion :

✓ Speed Unit Conversions

1. Convert km/hr to m/sec:

$$1 \text{ km/hr} = \frac{5}{18} \text{ m/sec}$$

Multiply by $\frac{5}{18}$.

- Example: $72 \text{ km/hr} = 72 \times \frac{5}{18} = 20 \text{ m/sec}$

2. Convert m/sec to km/hr:

$$1 \text{ m/sec} = \frac{18}{5} \text{ km/hr}$$

Multiply by $\frac{18}{5}$.

- Example: $10 \text{ m/sec} = 10 \times \frac{18}{5} = 36 \text{ km/hr}$

Relative Speed

When two objects move towards or away from each other, their speeds are combined.

✓ When Moving in Opposite Directions:

$$\text{Relative Speed} = \text{Speed}_1 + \text{Speed}_2$$

Example: Two trains moving towards each other at **40 km/hr** and **50 km/hr**

$$\text{Relative Speed} = 40 + 50 = 90 \text{ km/hr}$$

✓ When Moving in the Same Direction:

$$\text{Relative Speed} = |\text{Speed}_1 - \text{Speed}_2|$$

Example: A car at **60 km/hr** and another at **40 km/hr** in the same direction

$$\text{Relative Speed} = |60 - 40| = 20 \text{ km/hr}$$

Average Speed

✓ Different Distances, Same Time Taken:

$$\text{Average Speed} = \frac{S_1 + S_2}{2}$$

✓ Same Distance, Different Speeds:

$$\text{Average Speed} = \frac{2S_1S_2}{S_1 + S_2}$$

Example:

A car covers **half** the distance at **40 km/hr** and the other **half** at **60 km/hr**.

$$\text{Average Speed} = \frac{2 \times 40 \times 60}{40 + 60} = \frac{4800}{100} = 48 \text{ km/hr}$$

Height and Distance

✓ Line of Sight (LOS)

- The **imaginary line drawn from the observer's eye to the object** being viewed.



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✓ Angle of Elevation

- The **angle between the horizontal ground level and the line of sight when looking UP at an object**.
- Example: Looking at the top of a building from the ground.

✓ Angle of Depression

- The **angle between the horizontal ground level and the line of sight when looking DOWN at an object**.
- Example: A person looking at a car from the top of a tower.

✓ Height (Vertical Distance)

- The **perpendicular distance from the base to the top of an object**.
- Example: The height of a tower, tree, or building.

✓ Distance (Horizontal Distance)

- The horizontal distance between the observer and the base of the object.

2. Important Trigonometric Ratios

Angle (θ)	$\sin(\theta) = \text{Opposite} / \text{Hypotenuse}$	$\cos(\theta) = \text{Adjacent} / \text{Hypotenuse}$	$\tan(\theta) = \text{Opposite} / \text{Adjacent}$
0°	0	1	0
30°	$1/2$	$\sqrt{3}/2$	$1/\sqrt{3}$
45°	$1/\sqrt{2}$	$1/\sqrt{2}$	1
60°	$\sqrt{3}/2$	$1/2$	$\sqrt{3}$
90°	1	0	Undefined

✂ Key Formula:

$$\tan(\theta) = \frac{\text{Height}}{\text{Distance}}$$

$$\text{Height} = \text{Distance} \times \tan(\theta)$$

Formulas :

✓ Basic Formula using Tan θ

$$\text{Height} = \text{Distance} \times \tan(\theta)$$

✓ Two Heights in the Same Plane

If two different angles of elevation θ_1 and θ_2 are observed from the same point,

$$\text{Height} = \frac{d(\tan \theta_1 \tan \theta_2)}{\tan \theta_2 - \tan \theta_1}$$

✓ If Observer Moves Closer to Object

If an observer moves closer, and the angle of elevation changes from θ_1 to θ_2 ,

$$d = \frac{h}{\tan \theta_2 - \tan \theta_1}$$

✓ Two Observers at Different Distances

If two observers at distances D_1 and D_2 see an object at angles θ_1 and θ_2 ,

$$h = \frac{(D_2 - D_1) \times \tan \theta_1 \times \tan \theta_2}{\tan \theta_2 - \tan \theta_1}$$

Common Problems

✓ 1. Finding the Height of a Tower

Problem: A person standing 50 m away from a tower sees the top at an angle of elevation 30° . Find the tower's height.

Solution:

$$\text{Height} = \text{Distance} \times \tan(30^\circ)$$

$$h = 50 \times \frac{1}{\sqrt{3}}$$

$$h = \frac{50}{1.732} \approx 28.87 \text{ m}$$

✓ 2. Finding Distance of an Object

Problem: A tree is 40 m tall. A man sees the top at an angle of elevation 60° . Find the distance between the man and the tree's base.

Solution:

$$\text{Distance} = \frac{\text{Height}}{\tan(60^\circ)}$$

$$d = \frac{40}{\sqrt{3}} \approx 23.09 \text{ m}$$

✓ 3. Angle of Elevation Changes

Problem: A man at 200 m distance from a tower sees the top at 30° . He walks towards it, and the angle becomes 60° . Find the height of the tower.

Solution:

Using the formula:

$$d = \frac{h}{\tan(60^\circ) - \tan(30^\circ)}$$

Since $\tan(60^\circ) = \sqrt{3}$ and $\tan(30^\circ) = 1/\sqrt{3}$,

$$200 - d = \frac{h}{\sqrt{3} - 1/\sqrt{3}}$$

After solving, $h = 100\sqrt{3} \approx 173.2 \text{ m}$.

Tips

✓ 1. Use Tan for Most Problems

Most height and distance problems use:

$$\tan(\theta) = \frac{\text{Height}}{\text{Distance}}$$

✓ 2. Memorize Key Trigonometric Values

Remember sin, cos, and tan values for 30°, 45°, 60° to solve problems quickly.

✓ 3. Draw a Diagram

Always sketch the problem to visualize angles and distances.

✓ 4. Convert Angles Properly

Use degrees mode in calculators (avoid radian mode).

✓ 5. Practice Common Cases

- **Object Seen from a Distance** → Use $\tan \theta$.
- **Object Seen from Two Positions** → Use height-difference formulas.

Compound Interest



1. Key Concepts

✓ Principal (P)

- The **initial amount** of money invested or borrowed.

✓ Rate of Interest (R)

- The percentage at which the interest is calculated **per annum**.

✓ Time (T)

- The duration for which the interest is calculated (**in years**).

✓ Compound Interest (CI)

- Interest is **calculated on the principal and also on the accumulated interest** from previous periods.
- Unlike **Simple Interest (SI)**, where interest is only on the principal, CI grows **exponentially**.

Formulas

$$A = P \left(1 + \frac{R}{100} \right)^T$$

$$CI = A - P$$

Where:

- A = Final Amount
- P = Principal
- R = Rate of Interest (per annum in %)
- T = Time (in years)
- CI = Compound Interest

✓ 1. Interest Compounded Annually

$$A = P \left(1 + \frac{R}{100} \right)^T$$

✓ 2. Interest Compounded Half-Yearly

- Dividing the rate by 2 and doubling the time:

$$A = P \left(1 + \frac{R}{200} \right)^{2T}$$

✓ 3. Interest Compounded Quarterly

- Dividing the rate by 4 and multiplying the time by 4:

$$A = P \left(1 + \frac{R}{400} \right)^{4T}$$

✓ 4. Interest Compounded Monthly

- Dividing the rate by 12 and multiplying the time by 12:

$$A = P \left(1 + \frac{R}{1200} \right)^{12T}$$

4. Difference Between Simple & Compound Interest

Factor	Simple Interest (SI)	Compound Interest (CI)
Formula	$SI = \frac{P \times R \times T}{100}$	$CI = P \left(1 + \frac{R}{100}\right)^T - P$
Growth Type	Linear Growth	Exponential Growth
Interest Applies On	Only on Principal	On Principal + Previous Interest
Returns Over Time	Lower than CI	Higher than SI

Imporant Formuals :

✓ 1. Compound Interest for 2 Years

$$CI = P \left[\left(1 + \frac{R}{100}\right)^2 - 1 \right]$$

OR

$$CI = P \left(\frac{2R}{100} + \frac{R^2}{100^2} \right)$$

✓ 2. Compound Interest for 3 Years

$$CI = P \left[\left(1 + \frac{R}{100}\right)^3 - 1 \right]$$

OR

$$CI = P \left(\frac{3R}{100} + \frac{3R^2}{100^2} + \frac{R^3}{100^3} \right)$$

✓ 3. Doubling & Tripling Money Using Compound Interest

- To find time (T) when money doubles:

$$T = \frac{\log 2}{\log(1 + R/100)}$$

- To find time (T) when money triples:

$$T = \frac{\log 3}{\log(1 + R/100)}$$

✓ 4. Rule of 72 (Approximate Calculation for Doubling Money)

- Time required to double an investment

$$T = \frac{72}{R}$$

- Example: If R = 8%,

$$T = \frac{72}{8} = 9 \text{ years (approximately)}$$

Tips :

- ✓ 1. If interest is compounded annually, use the standard formula:

$$A = P \left(1 + \frac{R}{100} \right)^T$$

- ✓ 2. If interest is compounded half-yearly, divide rate by 2 and multiply time by 2.

$$A = P \left(1 + \frac{R}{200} \right)^{2T}$$

- ✓ 3. If interest is compounded quarterly, divide rate by 4 and multiply time by 4.

$$A = P \left(1 + \frac{R}{400} \right)^{4T}$$

- ✓ 4. Use the "Rule of 72" for quick calculations to determine when money will double.

$$T = \frac{72}{R}$$

- ✓ 5. Always compare SI vs CI for better understanding of profit growth.

- ✓ 6. If compound interest is given for 2 or 3 years, use shortcut formulas:

$$CI = P \left(\frac{2R}{100} + \frac{R^2}{100^2} \right) \quad (\text{for 2 years})$$

$$CI = P \left(\frac{3R}{100} + \frac{3R^2}{100^2} + \frac{R^3}{100^3} \right) \quad (\text{for 3 years})$$



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Common Problems :

✓ 1. Find Compound Interest Over 2 Years

Problem:

A sum of ₹10,000 is invested at 10% per annum compounded annually for 2 years. Find the compound interest.

Solution:

Using CI Formula for 2 years:

$$A = 10,000 \times \left(1 + \frac{10}{100} \right)^2$$

$$A = 10,000 \times (1.1)^2 = 10,000 \times 1.21 = 12,100$$

$$CI = A - P = 12,100 - 10,000 = 2,100$$

✓ 2. Find Compound Interest When Compounded Half-Yearly

Problem:

A sum of ₹20,000 is deposited at 8% per annum, compounded half-yearly for 1 year. Find the compound interest.

Solution:

Since interest is compounded half-yearly,

$$R = \frac{8}{2} = 4\% \quad \text{and} \quad T = 2 \quad (\text{since 1 year} = 2 \text{ half-years})$$

$$A = 20,000 \times \left(1 + \frac{4}{100} \right)^2$$

$$A = 20,000 \times (1.04)^2 = 20,000 \times 1.0816 = 21,632$$

$$CI = A - P = 21,632 - 20,000 = 1,632$$

Permutation and Combination

✔ Factorial (!)

Factorial of a number $n!$ is the product of all natural numbers from n down to 1.

$$n! = n \times (n - 1) \times (n - 2) \times \dots \times 1$$

- Example: $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$
- Special Cases:
 - $0! = 1$ (By definition)
 - $1! = 1$

✔ Permutation (Ordered Selection)

A permutation is an arrangement of objects in a specific order.

$$P(n, r) = \frac{n!}{(n - r)!}$$

Where:

- n = Total number of objects
- r = Number of objects to arrange
- Order matters

Example:

How many ways can you arrange 3 letters out of 5 (A, B, C, D, E)?

$$P(5, 3) = \frac{5!}{(5 - 3)!} = \frac{5!}{2!} = \frac{5 \times 4 \times 3 \times 2!}{2!} = 5 \times 4 \times 3 = 60$$

✔ Combination (Unordered Selection)

A combination is a selection of objects where order does NOT matter.

$$C(n, r) = \frac{n!}{r!(n - r)!}$$

Where:

- n = Total number of objects
- r = Number of objects to choose
- Order does NOT matter

Example:

How many ways can you select 3 letters out of 5 (A, B, C, D, E)?

$$C(5, 3) = \frac{5!}{3!(5 - 3)!} = \frac{5 \times 4 \times 3!}{3! \times 2!} = \frac{5 \times 4}{2!} = \frac{20}{2} = 10$$

Difference

Feature	Permutation (P)	Combination (C)
Order	Matters	Doesn't Matter
Formula	$P(n, r) = \frac{n!}{(n-r)!}$	$C(n, r) = \frac{n!}{r!(n-r)!}$
Example	Arranging 3 books in a row	Choosing 3 books from a shelf
Real-life Example	Seating arrangements, Passwords	Lottery selection, Committee formation

✓ 1. When $n = r$ (All objects are used)

- Permutation: $P(n, n) = n!$
- Combination: $C(n, n) = 1$ (Only one way to choose all objects)

✓ 2. Choosing None ($r = 0$)

- Permutation: $P(n, 0) = 1$
- Combination: $C(n, 0) = 1$

✓ 3. Choosing 1 Object ($r = 1$)

- Permutation: $P(n, 1) = n$
- Combination: $C(n, 1) = n$

✓ 4. When $r = n - 1$

$$C(n, n - 1) = C(n, 1) = n$$

✓ 5. Symmetric Property of Combinations

$$C(n, r) = C(n, n - r)$$

Example:

$$C(5, 2) = C(5, 3) = 10$$

Common Problems

✓ 1. Arranging People in a Line (Permutation Problem)

Problem:

How many ways can 4 people be arranged in a row?

Solution:

Since order matters, use permutation:

$$P(4, 4) = 4! = 4 \times 3 \times 2 \times 1 = 24$$

✓ 2. Selecting a Committee (Combination Problem)

Problem:

A company needs to select 3 members from a group of 7 employees. In how many ways can they be selected?

Solution:

Since order does not matter, use combination:

$$C(7, 3) = \frac{7!}{3!(7-3)!} = \frac{7 \times 6 \times 5}{3 \times 2 \times 1} = \frac{210}{6} = 35$$

✓ 3. Arranging Digits in a Number (Permutation Problem)

Problem:

How many 3-digit numbers can be formed using the digits 1, 2, 3, 4, 5 without repetition?

Solution:

Since order matters, use permutation:

$$P(5, 3) = \frac{5!}{(5-3)!} = \frac{5!}{2!} = 5 \times 4 \times 3 = 60$$

✓ 4. Choosing a Cricket Team (Combination Problem)

Problem:

A cricket team of 11 players needs to be selected from 16 available players. How many ways can this be done?

Solution:

Since order does not matter, use combination:

$$C(16, 11) = \frac{16!}{11!(16-11)!} = \frac{16 \times 15 \times 14 \times 13 \times 12}{5!} = 4368$$

5. Special Cases in Permutations & Combinations

✓ 1. Arranging Objects with Repetition

If a word contains repeated letters, divide by the factorial of those repeated letters.

$$\text{Ways} = \frac{n!}{a! \times b! \times c!}$$

- Example: Arranging "AAB"

$$\frac{3!}{2!} = \frac{6}{2} = 3 \text{ ways}$$

✓ 2. Circular Permutations

- Linear arrangement: $n!$
- Circular arrangement: $(n - 1)!$

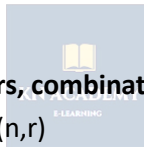
Example:

Seating 5 people around a circular table:

$$(5 - 1)! = 4! = 24$$

Tips

- ✓ 1. Use permutations when order matters, combinations when order does not.
- ✓ 2. Remember key formulas $P(n,r)$ and $C(n,r)$
- ✓ 3. Use factorial simplifications for faster calculations.
- ✓ 4. When dealing with repetition, divide by factorials of repeated items.
- ✓ 5. For circular arrangements, use $(n-1)!$ instead of $n!$
- ✓ 6. Apply the symmetric property: $C(n,r)=C(n,n-r)$
- ✓ 7. If given "at least" or "at most" cases, break into multiple calculations.



Probability

✓ 1. Probability Definition

Probability measures **how likely** an event is to occur.

Probability (P)=Favorable Outcomes/Total Outcomes

- The probability of an event **always** lies between **0 and 1**: $0 \leq P(E) \leq 1$
- $P(E) = 0 \rightarrow$ Impossible event
- $P(E) = 1 \rightarrow$ Certain event

✓ 2. Types of Events

◆ Sure Event

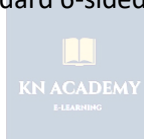
An event that **always** occurs.

- ◆ Example: Getting a number ≤ 6 when rolling a standard dice.

◆ Impossible Event

An event that **never** occurs.

- ◆ Example: Getting a 7 when rolling a standard 6-sided die.



◆ Mutually Exclusive Events

Two events that **cannot** happen at the same time.

- ◆ Example: Drawing a **King** and a **Queen** from a deck in one draw.

◆ Independent Events

Two events are independent if **one event does not affect the other**.

- ◆ Example: Tossing a coin **twice** (first toss doesn't affect the second).

◆ Dependent Events

Two events where **one event affects the probability of the other**.

- ◆ Example: Drawing two cards **without replacement** from a deck.

◆ Complementary Events

The **opposite** of an event happening.

$$P(\text{Not } A) = 1 - P(A)$$

- ♦ Example: If the probability of rain is **0.3**, the probability of **no rain** is **1 - 0.3 = 0.7**.

Formulas

1. Basic Probability Formula

$$P(E) = \frac{\text{Favorable Outcomes}}{\text{Total Outcomes}}$$

2. Probability of Complementary Events

$$P(A') = 1 - P(A)$$

Where $P(A')$ is the probability that event **A does not occur**.

3. Addition Rule of Probability

(i) For Mutually Exclusive Events

$$P(A \cup B) = P(A) + P(B)$$

- ♦ Example: Probability of drawing an **Ace** or a **King** from a deck:

$$P(Ace) + P(King) = \frac{4}{52} + \frac{4}{52} = \frac{8}{52}$$

(ii) For Non-Mutually Exclusive Events

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

- ♦ Example: Probability of drawing a **red card** or a **King** from a deck:

$$P(Red) + P(King) - P(Red \cap King)$$

Multiplication Rule

(i) For Independent Events

$$P(A \cap B) = P(A) \times P(B)$$

- ♦ Example: Probability of **getting heads twice** when tossing a coin:

$$P(HH) = P(H) \times P(H) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

(ii) For Dependent Events

$$P(A \cap B) = P(A) \times P(B|A)$$

- ♦ Example: Drawing **two Aces without replacement**:

$$P(A_1) = \frac{4}{52}, \quad P(A_2|A_1) = \frac{3}{51}$$

$$P(A_1 \cap A_2) = \frac{4}{52} \times \frac{3}{51} = \frac{12}{2652} = \frac{1}{221}$$

5. Bayes' Theorem (Conditional Probability)

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)}$$

- ♦ Used when probability depends on prior conditions.

Special Cases

✓ 1. Probability of At Least One Occurrence

$$P(\text{At least one}) = 1 - P(\text{None})$$

- ♦ Example: Probability of getting at least one head in two coin tosses:

$$P(\text{No heads}) = P(TT) = \frac{1}{4}$$

$$P(\text{At least one head}) = 1 - \frac{1}{4} = \frac{3}{4}$$

✓ 2. Probability of Repeated Independent Events

$$P(\text{Event happening at least once}) = 1 - (P(\text{Failure}))^n$$

- ♦ Example: Probability of rolling at least one 6 in 4 rolls of a die:

$$P(\text{Not rolling a 6}) = \frac{5}{6}$$

$$P(\text{At least one 6}) = 1 - \left(\frac{5}{6}\right)^4$$

Cards

- Total cards in a standard deck = 52
- Red cards = 26 (Diamonds & Hearts)
- Black cards = 26 (Clubs & Spades)
- Aces = 4, Kings = 4, Queens = 4, Jacks = 4

- ♦ Probability of drawing an Ace:

$$P(\text{Ace}) = \frac{4}{52} = \frac{1}{13}$$

- ♦ Probability of drawing a face card (King, Queen, Jack):

$$P(\text{Face Card}) = \frac{12}{52} = \frac{3}{13}$$

Tips

- ✓ 1. When you see "at least one," use:

$$P(\text{At least one}) = 1 - P(\text{None})$$

- ✓ 2. If order matters, check if it's a permutation problem.
- ✓ 3. If order doesn't matter, it's likely a combination problem.
- ✓ 4. Mutually exclusive events don't overlap, so use direct addition.
- ✓ 5. If dependent events, adjust the probability after each selection.
- ✓ 6. For card problems, know the deck structure well.
- ✓ 7. Probability values always range between 0 and 1. If you get >1 , it's incorrect.

Ratio

✓ 1. Ratio Definition

A **ratio** compares two or more numbers, showing how much of one value exists compared to another.

Ratio = First Quantity / Second Quantity

- ♦ Example: The ratio of **boys to girls** in a class is **3:5**, meaning for every 3 boys, there are 5 girls.



✓ 2. Proportion Definition

A **proportion** states that two **ratios** are equal.

$$a:b=c:d \text{ or } a/b = c/d$$

- ♦ Example: If 2 pens cost ₹10, then 4 pens should cost ₹20.

$$2:10=4:20$$

Types of Ratio

♦ 1. Duplicate Ratio

If the ratio is $a:b$, then the **duplicate ratio** is:

$$a^2 : b^2$$

- ♦ Example: If 3:5, then the duplicate ratio is 9:25.

♦ 2. Triplicate Ratio

If the ratio is $a:b$, then the **triplicate ratio** is:

$$a^3 : b^3$$

- ♦ Example: If 2:3, then the triplicate ratio is 8:27.

♦ 3. Sub-Duplicate Ratio

Square root of the given ratio:

$$\sqrt{a} : \sqrt{b}$$

- ♦ Example: If 9:16, then the sub-duplicate ratio is 3:4.

♦ 4. Sub-Triplicate Ratio

Cube root of the given ratio:

$$\sqrt[3]{a} : \sqrt[3]{b}$$

- ♦ Example: If 27:125, then the sub-triplicate ratio is 3:5.

♦ 5. Inverse Ratio

Flipping the given ratio:

$$b : a$$

- ♦ Example: The inverse ratio of 5:7 is 7:5.

Ratio Formulas

1. Basic Ratio Formula

$$\text{Ratio} = \frac{\text{First Quantity}}{\text{Second Quantity}}$$

2. Equivalent Ratios (Cross Multiplication)

If $a:b = c:d$, then:

$$a \times d = b \times c$$

- ♦ Example: If $4:6 = 6:9$, then:

$$4 \times 9 = 6 \times 6$$

3. If two numbers are in ratio $a:b$ and their sum is S , then the numbers are:

$$\frac{a}{a+b} \times S \quad \text{and} \quad \frac{b}{a+b} \times S$$

- ♦ Example: Two numbers are in ratio $3:5$ and their sum is 80 .

$$\frac{3}{3+5} \times 80 = 30, \quad \frac{5}{3+5} \times 80 = 50$$

4. If $a:b$ and $b:c$ are given, then $a:c$ is found by multiplying ratios:

$$\frac{a}{b} \times \frac{b}{c} = \frac{a}{c}$$

- ♦ Example: If $a:b = 3:4$ and $b:c = 5:6$, then:

$$a : c = (3/4) \times (5/6) = 15 : 24 \Rightarrow 5 : 8$$

5. If $a:b = c:d = e:f = k$, then:

$$\frac{a+c+e}{b+d+f} = k$$

Proportion Formulas

1. Direct Proportion (More-More Relationship)

If x increases, y also increases (e.g., salary vs. working hours), then:

$$\frac{x_1}{y_1} = \frac{x_2}{y_2}$$

- ♦ Example: If 4 workers take 8 hours, how many hours will 8 workers take?

$$\frac{4}{8} = \frac{8}{x} \Rightarrow x = 4$$

2. Inverse Proportion (More-Less Relationship)

If x increases, y decreases (e.g., speed vs. time), then:

$$x_1 y_1 = x_2 y_2$$

- ♦ Example: If 3 machines take 12 hours, how many hours will 6 machines take?

$$3 \times 12 = 6 \times x \Rightarrow x = 6$$

Tips

- ✓ 1. If $A:B = 2:3$ and $B:C = 4:5$, then $A:B:C$ can be found by multiplying corresponding values.

♦ Example:

$$A : B = 2 : 3, \quad B : C = 4 : 5$$

Multiply such that **B** matches in both ratios (LCM of 3 & 4 = 12):

$$A : B = (2 \times 4) : (3 \times 4) = 8 : 12$$

$$B : C = (3 \times 4) : (5 \times 3) = 12 : 15$$

Thus, $A:B:C = 8:12:15$.

- ✓ 2. If income and expense are in a ratio, use the unitary method to find savings.

♦ Example: If income:expense = 5:3, then savings are $5-3 = 2$ parts of total income.

- ✓ 3. When working with partnership problems, profit is divided based on:

$$\text{Profit Share} = \frac{\text{Investment} \times \text{Time}}{\text{Total Investment}}$$

♦ Example: A invests ₹10,000 for 1 year, and B invests ₹15,000 for 8 months. Find their profit ratio.

$$A : B = (10,000 \times 12) : (15,000 \times 8) = 120,000 : 120,000 = 1 : 1$$

- ✓ 4. Ratio Split for Mixtures

$$\text{New Ratio} = \frac{(\text{Quantity} \times \text{Existing Ratio}) + \text{Added Quantity}}{\text{Total Quantity}}$$

♦ Example: A milk-water solution is 4:1, and 2L of water is added to 10L of the solution. New ratio?

$$\text{New Water} = (1 \times 10) + 2 = 12$$

$$\text{New Milk} = 4 \times 10 = 40$$

$$\text{New Ratio} = 40 : 12 = 10 : 3$$

LCM & HCF

- ✓ 1. Highest Common Factor (HCF)

The **HCF (Greatest Common Divisor - GCD)** of two or more numbers is the **largest number** that divides all the given numbers **exactly**.

♦ Example: **HCF of 12 and 18**

Factors of **12** = {1, 2, 3, 4, 6, 12}

Factors of **18** = {1, 2, 3, 6, 9, 18}

Common factors = {1, 2, 3, 6}

HCF = 6

-
- ✓ 2. Lowest Common Multiple (LCM)

The **LCM** of two or more numbers is the **smallest number** that is **divisible by all** the given numbers.

◆ Example: **LCM of 4 and 6**

Multiples of 4 = {4, 8, 12, 16, 20, ...}

Multiples of 6 = {6, 12, 18, 24, ...}

Smallest common multiple = 12

LCM = 12

📖 Methods to Find HCF & LCM

Method 1: Prime Factorization Method

1. Find the **prime factors** of each number.
2. **HCF** = Product of **smallest power** of common prime factors.
3. **LCM** = Product of **highest power** of all prime factors.

◆ Example: **Find HCF & LCM of 24 and 36**

- $24 = 2^3 \times 3$

- $36 = 2^2 \times 3^2$

👉 **HCF = $2^2 \times 3 = 12$**

👉 **LCM = $2^3 \times 3^2 = 72$**

Method 2: Division Method (For HCF Only)



1. Divide the **larger number** by the **smaller number**.
2. Take the remainder and divide it into the previous divisor.
3. Repeat the process until the remainder is **0**.
4. The last divisor is the **HCF**.

◆ Example: **Find HCF of 48 and 72**

- $72 \div 48 = 1$, remainder = 24

- $48 \div 24 = 2$, remainder = 0

👉 **HCF = 24**

Method 3: Listing Multiples Method (For LCM Only)

1. Write multiples of each number.
2. Find the smallest **common multiple**.

◆ Example: **Find LCM of 5 and 7**

- Multiples of 5: {5, 10, 15, 20, 25, 30, 35, ...}

- Multiples of 7: {7, 14, 21, 28, 35, 42, ...}

👉 LCM = 35

📌 Key HCF & LCM Formulas

1. HCF × LCM Formula

$$HCF(a, b) \times LCM(a, b) = a \times b$$

- Example: If $HCF(8, 12) = 4$, find LCM.

$$4 \times LCM = 8 \times 12$$

$$LCM = \frac{8 \times 12}{4} = 24$$

2. LCM of Fractions

$$LCM = \frac{\text{LCM of Numerators}}{\text{HCF of Denominators}}$$

3. HCF of Fractions

$$HCF = \frac{\text{HCF of Numerators}}{\text{LCM of Denominators}}$$

- Example: Find HCF & LCM of $\frac{3}{4}$ and $\frac{5}{6}$

- HCF of 3, 5 = 1
- LCM of 4, 6 = 12
- LCM of $\frac{3}{4}$ & $\frac{5}{6} = \frac{(3 \times 5)}{2} = \frac{15}{2}$
- HCF of $\frac{3}{4}$ & $\frac{5}{6} = \frac{1}{12}$

- ♦ Example: Find HCF & LCM of $\frac{3}{4}$ and $\frac{5}{6}$

- HCF of 3, 5 = 1
- LCM of 4, 6 = 12
- LCM of $\frac{3}{4}$ & $\frac{5}{6} = \frac{(3 \times 5)}{2} = \frac{15}{2}$
- HCF of $\frac{3}{4}$ & $\frac{5}{6} = \frac{1}{12}$

📌 Special HCF & LCM Tricks

- ✓ 1. If two numbers are co-prime, their HCF is always 1.
- ✓ 2. LCM of consecutive numbers is their product.
- ✓ 3. If two numbers have the same HCF and LCM, they are equal.

- ✓ 4. HCF of two even numbers is always even.
- ✓ 5. HCF of two odd numbers is always odd.

5.5 Shortcut Questions & Examples

Example 1: Find HCF and LCM of 18 and 24.

- **Factors of 18** = {1, 2, 3, 6, 9, 18}
- **Factors of 24** = {1, 2, 3, 4, 6, 8, 12, 24}
- **HCF** = 6
- **LCM** = 72

Example 2: Find the largest number that exactly divides 72 and 108.

- $\text{HCF}(72, 108) = 36$
 👉 Answer = 36

Example 3: Two numbers have LCM 120 and HCF 10. If one number is 40, find the other.

Using **HCF × LCM = Product of Numbers**,

$$10 \times 120 = 40 \times x \quad 10 \times 120 = 40 \times x \quad x = \frac{10 \times 120}{40} = 30$$

👉 Answer = 30

Example 4: Find LCM of 8, 12, and 16.

- **Prime Factorization:**
 - $8 = 2^3$
 - $12 = 2^2 \times 3$
 - $16 = 2^4$
- **LCM** = $2^4 \times 3 = 48$
 👉 Answer = 48

Example 5: The LCM and HCF of two numbers are 96 and 8. One number is 32. Find the other.

$\text{HCF} \times \text{LCM} = \text{Product of Two Numbers}$

$$8 \times 96 = 32 \times x$$

$$x = \frac{8 \times 96}{32} = 24$$

👉 Answer = 24

6 Applications of HCF & LCM

♦ LCM Applications:

- ✓ Finding the **least time interval** for repeated events.
- ✓ Finding the **least number of objects needed** to form groups.

♦ HCF Applications:

- ✓ Distributing items into **equal groups**.
 - ✓ Finding the **maximum possible length** for cutting materials.
-

7 Quick Recap of Formulas

Concept	Formula
HCF × LCM	$\text{HCF} \times \text{LCM} = \text{Product of Numbers}$
LCM of Fractions	LCM of Numerators/HCF of Denominators
HCF of Fractions	HCF of Numerators/LCM of Denominators

